

Python Intermediate Day 3 Checkpoint

- Operator Overloading
- Designing with Classes
- Advanced Class Topics
- Decorators & Class Gotchas
- Software Engineering term
- Best practice of development with LLM

1. Email *

2. Name

3. If you want to support the + operator for a custom class, you must implement:

1 point

Mark only one oval.

- ☐ A. __plus__
- ☐ B. __add__
- ☐ C. __sum__
- ☐ D. __concat__

4. Which method enables indexing like `obj[0]`?

1 point

Mark only one oval.

- ☐ A. `__index__`
- ☐ B. `__getitem__`
- ☐ C. `__access__`
- ☐ D. `__slice__`

5. To override the `<` operator, you implement:

1 point

Mark only one oval.

- ☐ A. `__less__`
- ☐ B. `__lt__`
- ☐ C. `__cmp__`
- ☐ D. `__compare__`

6. What is the main advantage of composition over inheritance?

1 point

Mark only one oval.

- ☐ A. Faster execution
- ☐ B. Less memory usage
- ☐ C. Better flexibility and loose coupling
- ☐ D. Automatic polymorphism

7. “Is-a” relationship represents:

1 point

Mark only one oval.

- ☐ A. Composition
- ☐ B. Delegation
- ☐ C. Inheritance
- ☐ D. Factory pattern

8. Why is an Abstract Base Class (ABC) used in Python?

1 point

Mark only one oval.

- ☐ A. To improve program speed
- ☐ B. To store abstract data types
- ☐ C. To enforce that subclasses implement required methods
- ☐ D. To prevent inheritance

9. What happens if you try to instantiate an abstract class that contains an unimplemented `@abstractmethod`?

1 point

Mark only one oval.

- ☐ A. The object is created successfully
- ☐ B. Python automatically implements the method
- ☐ C. A `TypeError` is raised
- ☐ D. The program ignores the abstract method

10. Which special method allows an object to be used in a for loop?

1 point

Mark only one oval.

- ☐ A. `__len__()`
- ☐ B. `__getitem__()`
- ☐ C. `__iter__()`
- ☐ D. `__contains__()`

11. What is the purpose of Python docstrings?

1 point

Mark only one oval.

- ☐ A. To increase execution speed
- ☐ B. To automatically fix coding errors
- ☐ C. To document classes, methods, and functions for users and developers
- ☐ D. To replace comments completely

12. What is hallucination in generative AI?

1 point

Mark only one oval.

- ☐ A. When an AI system shuts down
- ☐ B. When an AI generates information that is inaccurate or unsupported
- ☐ C. When an AI becomes self-aware
- ☐ D. When an AI loses internet access

13. In AI systems, what is a **token**?

1 point

Mark only one oval.

- ☐ A. A programming language keyword only
- ☐ B. A unit of text, such as a word, subword, or punctuation, that the AI processes
- ☐ C. A type of computer processor
- ☐ D. A software license key

14. Which feature allows attribute validation logic?

1 point

Mark only one oval.

- ☐ A. Mixins
- ☐ B. Descriptors / Properties
- ☐ C. MRO
- ☐ D. Metaclasses

15. In UML, inheritance is represented by:

1 point

Mark only one oval.

- ☐ A. Dashed arrow
- ☐ B. Solid line with hollow triangle
- ☐ C. Double arrow
- ☐ D. Diamond shape

16. In object design, “cross-cutting concerns” include:

1 point

Mark only one oval.

- ☐ A. Data structures
- ☐ B. Logging and authentication
- ☐ C. Inheritance trees
- ☐ D. Class variables

17. Why prefer getters and setters (or properties)?

1 point

Mark only one oval.

- ☐ A. Slower execution
- ☐ B. Direct attribute access is unsafe
- ☐ C. Encapsulation and validation control
- ☐ D. Required by Python

18. **What is the purpose of a system prompt?**

1 point

Mark only one oval.

- ☐ A. To establish instructions or behavior for the AI model
- ☐ B. To increase computer RAM
- ☐ C. To encrypt a hard drive
- ☐ D. To connect a printer

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